

Randomization

19/02/2024

Description

This file enables concatenation and randomization of sign strings which occur on different parts of an object (indicated by underscore '_'). We do not know the order of these strings (if any), hence, if we want to combine them, they need to be randomized.

Session Info

Give the session info (reduced).

```
## [1] "R version 3.6.3 (2020-02-29)"
## [1] "x86_64-pc-linux-gnu"
```

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
#library(stringr)
```

Load Data

```
sign.data <- read.csv("~/Github/UpperPalCombinatorics/data/signBase/signBase_clean.csv")
head(sign.data)
```

```
##   object_id ensemble_id archaeological_unit      site_name location
## 1 sgr0074   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière    NA
## 2 sgr0073   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière    NA
## 3 sgr0072   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière    NA
## 4 sgr0070   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière    NA
## 5 sgr0067   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière    NA
## 6 sgr0062   sgr_ens_01 Middle Magdalenian Saint-Germain-la-Rivière    NA
##   layer      dating_method date_bp_min min_lab_number date_bp_max
## 1 burial AMS, 858 (Gifa 95456) 15,780 ± 200          NA          NA
## 2 burial AMS, 858 (Gifa 95456) 15,780 ± 200          NA          NA
## 3 burial AMS, 858 (Gifa 95456) 15,780 ± 200          NA          NA
## 4 burial AMS, 858 (Gifa 95456) 15,780 ± 200          NA          NA
## 5 burial AMS, 858 (Gifa 95456) 15,780 ± 200          NA          NA
## 6 burial AMS, 858 (Gifa 95456) 15,780 ± 200          NA          NA
```

```
## max_lab_number excavation_year material
## 1 NA 1934 animal tooth, red deer
## 2 NA 1934 animal tooth, red deer
## 3 NA 1934 animal tooth, red deer
## 4 NA 1934 animal tooth, red deer
## 5 NA 1934 animal tooth, red deer
## 6 NA 1934 animal tooth, red deer
## object_type length_mm width_mm depth_mm preservation
## 1 personal ornament, pendant NA NA NA complete
## 2 personal ornament, pendant NA NA NA complete
## 3 personal ornament, pendant NA NA NA complete
## 4 personal ornament, pendant NA NA NA fragmentary
## 5 personal ornament, pendant NA NA NA fragmentary
## 6 personal ornament, pendant NA NA NA complete
## sign_coding coding_certainty coding_clean
## 1 |||X Certain |||X
## 2 || Certain ||
## 3 XX Very uncertain XX
## 4 |||| Certain ||||
## 5 || Very uncertain ||
## 6 ||| Certain |||
```

Randomize

Randomize sign sequences per object.

```
# set seed to get reproducible result
set.seed(09012020)
# initialize empty vector
coding.random <- c()
# run loop to randomize, and add new column to data frame
for (i in 1:nrow(sign.data)){
  # use single coding per row
  coding.local <- sign.data[i, ]$coding_clean
  # split on underscores surrounded by white spaces
  coding.split <- unlist(strsplit(as.character(coding.local), split = " _ "))
  # randomize order
  coding.random.local <- sample(coding.split)
  # concatenate with underscore and white space again
  coding.random.local <- paste(coding.random.local, collapse = " _ ")
  # add to vector
  coding.random <- c(coding.random, coding.random.local)
}

# add column with randomized order to data frame
sign.data$coding_random <- coding.random

# save to file
write.csv(sign.data, "~/Github/UpperPalCombinatorics/data/signBase/signBase_randomized.csv", row.names = FALSE)
```